Notes on Ground Checking & Jumping: 3/24/26

- I created the BoxCastGrounded.cs Script to check if the player's vertical raycast is colliding with the ground.
- The raycast itself appears to be working from the color changing of the line.
- However, I need to update this value accordingly inside of PlayerMovement.cs which receives the input for jumping.

- The Physics.Raycast() method casts a ray from the point "origin" in the direction "direction" of length "maxDistance" against <u>all</u> colliders in a scene.

- There's a couple different parameter configurations.

```
Raycast (Ray ray, out RaycastHit hitInfo, float maxDistance)
```

One of them is just this

Another one is all four.

- However, the docs specify one with yet <u>another</u> variable, specifically for layer masks.

- The layerMask parameter is used to selectively filter which colliders are considered when casting a ray, which I'm thinking could be good for preventing unexpected errors and also optimizing the code.

- I was able to add this in the BoxCastGrounded.cs Script in the FixedUpdate() method, where the hittingGround boolean is located.

```
hittingGround = Physics.Raycast(groundInteractionRay,
                                out groundRaycastHit,
                                groundRayLength,
                                layerMask)
```

- When the raycast hits an object in the "Ground" layer (passed in through layerMask), it registers the change by setting hittingGround to true.

- There's also a public static event named groundCheck which carries a bool value.

- We fire the event once per FixedUpdate() after the if-else statement regarding the hittingGround bool.

- The bool is, of course, packaged up with the firing of this event, which is later picked up by PlayerMovement.cs.

- There might be a more optimal way to perform these checks for the raycast, but for right now it's working.